

SDG Goal 12

Responsible consumption and production

SDG Target 12.2

By 2030, achieve the sustainable management and efficient use of natural resources

SDG Indicator 12.2.1

Material footprint, material footprint per capita, and material footprint per GDP

Time series

Raw material consumption (RMC)

1. General information on the time series

- Date of national metadata: 7 October 2022
- National data: <http://sdg-indicators.de/12-2-1/>
- Definition: The time series measures the amount of raw material used globally for the production of goods for domestic consumption, gross fixed capital formation, changes in inventories and acquisitions less disposals of valuables. The time series is also available in relation to the domestic population and real gross domestic product (GDP).
- Disaggregation: Not available.

2. Comparability with the UN metadata

- Date of UN metadata: March 2025
- UN metadata: <https://unstats.un.org/sdgs/metadata/files/Metadata-12-02-01.pdf>
- The time series is not compliant with the UN metadata, but provides additional information. The RMC calculation is based on a hybrid model using domestic input-output-tables, life cycle information and other data sources. In the UN metadata, a multi-regional input-output (MRIO) framework is mentioned. Further methodological differences may exist e.g. due to deviations in the treatment of secondary raw materials.

3. Data description

- The data on raw material consumption is generated based on a hybrid estimation model combining physical and monetary information from official and other data sources. These calculations are project work run by the Environmental-Economic Accounts unit of the Federal Statistical Office.

The population data comes from the intercensal population updates, the basis of which is the last census conducted in 2011. The population data is rolled forward using statistical results on natural population change (births, deaths) and migrations. For 2010, the population was calculated backwards using the 2011 census and migration, birth and death statistics.

The data on GDP is calculated by the Federal Statistical Office's National Accounts as a secondary statistic. GDP is adjusted based on a price base changing every year (previous year's price base). After several revisions due to new data input, final results are available four years after the first preliminary release.

4. Access to data source

- Statistical Report – Raw material equivalents (RMC) (only available in German):
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/UGR/rohstoffe-materialfluesse-wasser/_inhalt.html
- National accounts – Gross value added, gross domestic product (nominal/price-adjusted) – GENESIS online 81000-0001:
<https://www-genesis.destatis.de/genesis//online?operation=table&code=81000-0001&bypass=true&language=en>
- Average population – GENESIS online 12411-0041:
<https://www-genesis.destatis.de/genesis//online?operation=table&code=12411-0041&bypass=true&levelindex=1&levelid=1639396599054>
- Population data based on Census 2011 – 1991 to 2011 (only available in German):
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Bevoelkerung/Bevoelkerungsstand/_inhalt.html

5. Metadata on source data

- Raw materials for Germany - analysis of consumption, investment and export at the macro and meso levels (only available in German):
<https://www.umweltbundesamt.de/publikationen/rohstoffe-fuer-deutschland>
- Quality Report – Microcensus (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Bevoelkerung/einfuehrung.html>
- Quality report – Economic Accounts (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Umwelt/ugr.pdf?>
- Project report: Improving Raw Material Equivalent Estimation in Environmental-Economic Accounts:
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/UGR/rohstoffe-materialfluesse-wasser/Publikationen/Downloads/improving-raw-material-equivalents-2025-1_2025028.pdf

6. Timeliness and frequency

- Timeliness: Irregular
- Frequency: Annual

7. Calculation method

- Unit of measurement: Million tonnes; Tonnes per capita
- Calculation:

RMC : calculation based on a hybrid model

$$\text{RMC [Mn t]} = \frac{\text{RMC [metric t]}}{1,000,000}$$

$$\text{RMC [t per capita]} = \frac{\text{RMC [metric t]}}{\text{Average population [number]}}$$

$$\text{RMC per real GDP [2015 = 100]} = \frac{\text{RMC}_{t1} [\text{metric t}] \cdot 100 [\%] / \text{RMC}_{t0} [\text{metric t}]}{\text{Real GDP [2015 = 100]}} \cdot 100 [\%]$$

$t1$ = current year; $t0$ = base year (2015)